

SymC in Oncology: Near-Critical Stability Control in Tumor–Therapy Dynamics

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Abstract

Revision Addendum: Version 3 update consolidates the framework by sharpening the justification for the adaptive χ -window, clarifying the underlying control structure, and adding explicit falsification criteria that strengthen the model’s scientific and clinical coherence.

This paper establishes clinical oncology as a testbed for the Symmetrical Convergence (SymC) framework’s near-critical damping principle. Building on theoretical validation across quantum, cosmological, and neurological systems, the analysis demonstrates that optimal cancer control operates in the adaptive window $0.8 \leq \chi \leq 1.0$, where $\chi \equiv \gamma/(2|\omega|)$ represents the damping ratio of tumor–immune–therapy dynamics. The same χ boundary appears in open quantum systems, cosmological structure formation, and neural control, where an information-efficiency functional $\eta(\chi)$ is maximized at $\chi = 1$ and remains high within 0.8–1.0. The CAT–OCT (Comparative Adaptive Therapy–Optimal Control Trial) protocol is introduced, using real-time circulating tumor DNA (ctDNA) kinetics to estimate $\chi(t)$ and guide dose modulation. By reframing cancer management as restoration of inherited near-critical stability, a substrate property propagated through biological organization, SymC provides a predictive, falsifiable alternative to the conventional Maximum Tolerated Dose paradigm. Explicit falsification criteria, observability requirements, and a translational roadmap from retrospective validation to Phase II trials are also presented.

1 Introduction

1.1 The universal near critical boundary

The same χ boundary appears in open quantum systems, cosmological structure growth, and neural control, where an information efficiency functional $\eta(\chi)$ is maximized at $\chi = 1$ and remains high in $0.8 \leq \chi \leq 1.0$. This universality traces to substrate inheritance: stable adaptive systems at each organizational scale operate near critically because they are built from substrates that themselves satisfy $\chi \approx 1$ boundary conditions. The constraint propagates upward from the cosmological bounce through symmetry breaking, QCD confinement, molecular assembly, and cellular organization.

Oncology thus provides a clinically accessible testbed of the same near critical stability principle observed at microscopic and cosmological scales. Cancer represents a breakdown of inherited $\chi \approx 1$ structure, with cells drifting into unstable underdamped regimes where uncontrolled proliferation dominates. Therapy aims to restore near critical dynamics, reestablishing the information efficient control that characterizes healthy tissue homeostasis.

1.2 From maximum tolerated dose to adaptive control

Conventional oncology seeks the Maximum Tolerated Dose (MTD): deliver the highest drug concentration patients can withstand, assuming monotonic dose–response relationships. This paradigm ignores temporal dynamics, feedback delays, and evolutionary pressures that drive resistance. Adaptive therapy [6] recognizes cancer as an evolving system and modulates treatment intensity based on real time burden estimates, trading maximal kill rates for long term control.

The SymC framework extends adaptive therapy by providing a quantitative control target: maintain the damping ratio $\chi(t)$ within $[0.8, 1.0]$. This window balances rapid suppression (with χ near 1 eliminating oscillations) with adaptive capacity (with χ slightly below 1 preserving information flow for parameter updates). Operating exactly at $\chi = 1$ achieves fastest monotonic convergence but eliminates feedback reserve; $\chi \approx 0.9$ optimizes the trade off between speed and robustness under realistic measurement noise and delays.

1.3 Structure of this paper

Section 2 derives χ from a minimal two state tumor control model and defines the stability objective. Section 3 operationalizes the adaptive window through the CAT–OCT protocol. Section 4 addresses observability and real time $\chi(t)$ estimation from ctDNA kinetics. Section 5 presents explicit falsification criteria. Section 6 maps the translational roadmap and acknowledges current limitations.

2 Mechanistic model and stability objective

2.1 Two state tumor control dynamics

Consider a minimal linearized model near an equilibrium state. Let:

- $X(t)$: tumor burden (for example total cell count or ctDNA concentration),
- $Y(t)$: effective drug concentration or immune activity,
- $u(t)$: therapeutic control input (dose rate).

Linearizing around equilibrium (X_0, Y_0) yields

$$\dot{X} = a_{11} (X - X_0) + a_{12} (Y - Y_0), \quad (1)$$

$$\dot{Y} = a_{21} (X - X_0) + a_{22} (Y - Y_0) + b u(t), \quad (2)$$

where a_{ij} encode net proliferation, immune suppression, and drug effect rates. The system Jacobian is

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}. \quad (3)$$

2.2 Characteristic frequencies and damping

The eigenvalues of A determine stability,

$$\lambda_{\pm} = -\gamma \pm i\omega, \quad (4)$$

with

$$\gamma = -\frac{1}{2} \text{Tr}(A) = -\frac{1}{2}(a_{11} + a_{22}), \quad (5)$$

$$\omega^2 = \det(A) - \gamma^2. \quad (6)$$

The damping ratio is defined as

$$\chi \equiv \frac{\gamma}{2|\omega|}. \quad (7)$$

Physical interpretation:

- $\gamma > 0$: effective decay rate of perturbations in tumor burden,
- ω : dominant oscillatory frequency extracted from biomarker dynamics,
- $\chi < 1$: underdamped, with oscillatory transients and risk of relapse peaks,
- $\chi = 1$: critically damped, with fastest monotonic approach to equilibrium,
- $\chi > 1$: overdamped, with slow response and loss of observability.

2.3 The adaptive window: $0.8 \leq \chi \leq 1.0$

While $\chi = 1$ marks the mathematical critical boundary, optimal clinical performance occurs near $\chi \approx 0.9$:

1. **Information efficiency:** slight underdamping enhances parameter observability from noisy ctDNA signals, improving adaptive controllers.
2. **Robustness to delays:** sampling lags and pharmacokinetic delays make exact $\chi = 1$ targeting fragile; the window $[0.8, 1.0]$ provides margin.
3. **Resistance management:** modest oscillations prevent complete quiescence of resistant subclones, maintaining evolutionary balance.

2.4 Control objective

The stability objective for parameter estimation or model predictive controller design is

$$J = \mathbb{E}[(\chi(t) - 0.9)^2] + \lambda_u \mathbb{E}[u(t)^2], \quad (8)$$

where λ_u balances efficacy and toxicity.

In practice, this corresponds to a constrained model predictive control (MPC) problem

$$\min_{u(\cdot)} J \quad (9)$$

subject to

$$u_{\min} \leq u(t) \leq u_{\max}, \quad (10)$$

$$\text{pharmacokinetic and pharmacodynamic constraints,} \quad (11)$$

$$\text{measurement and actuation delays.} \quad (12)$$

The CAT-OCT protocol implements this via simple guardrail rules (Section 3) that approximate the MPC solution while remaining clinically interpretable.

2.5 Symbol table

Symbol	Definition
$X(t)$	Tumor burden (cell count or surrogate biomarker)
$Y(t)$	Effective control state (drug or immune activity)
$u(t)$	Therapeutic control input (dose rate)
γ	Effective damping coefficient (decay rate)
ω	Characteristic oscillation frequency
$\chi(t)$	Damping ratio, $\gamma/(2 \omega)$
λ_u	Control cost regularization parameter
χ_{opt}	Optimal target damping ratio ≈ 0.9

3 CAT–OCT protocol: adaptive control in practice

The CAT–OCT protocol operationalizes the adaptive window through measurable biomarkers such as ctDNA, tumor markers (for example CA–125, PSA), or imaging derived burden estimates. For each patient, $\chi(t)$ is estimated from recent data via system identification methods (Section 4).

3.1 Operational guardrail rules

The dosing guardrails are:

$$\left\{ \begin{array}{ll} \chi(t) < 0.8 : & \text{Underdamped instability. Increase control intensity or shorten dosing interval.} \\ 0.8 \leq \chi(t) \leq 1.0 : & \text{Adaptive optimum. Maintain current regimen and monitor.} \\ 1.0 < \chi(t) \leq 1.2 : & \text{Acceptable margin. Observe for overdamping drift.} \\ \chi(t) > 1.2 : & \text{Overdamped rigidity. Reduce dose to restore adaptability.} \end{array} \right. \quad (13)$$

These quantitative thresholds define actionable guardrails for CAT–OCT dosing schedules, preserving feedback responsiveness while minimizing relapse probability.

3.2 Clinical workflow

A practical CAT–OCT workflow is:

1. **Baseline assessment:** establish X_0 and Y_0 via imaging and biomarker panel.
2. **Initial dosing:** begin with standard of care regimen.
3. **Monitoring phase:** collect ctDNA samples every 3–7 days (see observability requirements in Section 4.1).
4. **$\chi(t)$ estimation:** apply Prony or subspace methods to extract γ and ω from the recent biomarker trajectory.
5. **Decision rule:** adjust $u(t)$ according to the guardrail rules above.
6. **Iteration:** repeat steps 3–5 throughout the treatment course.

3.3 Resistance and multi state extensions

This two state formulation is a minimal model. In reality, tumor populations comprise sensitive, stem like, and resistant subclones. Future work will extend CAT–OCT to multi substrate models where each population has distinct χ_k values, and therapy maintains synchronization across substrates to prevent evolutionary escape. The current protocol targets aggregate $\chi(t)$ as a first order control variable.

4 Data requirements and real time estimation

4.1 Observability and identifiability

Real time $\chi(t)$ estimation depends on signal quality from ctDNA kinetics or related biomarkers. In a two state linear approximation with scalar measurement

$$z(t) = c_X X(t) + c_Y Y(t) + \text{noise}, \quad (14)$$

local observability requires:

1. **Sampling frequency:** interval Δt shorter than the characteristic approach time $1/\gamma$ (typically $\Delta t \lesssim 7$ days for solid tumors).
2. **Signal to noise ratio:** measurement coefficient of variation below about 20% to track $\chi(t)$ within ± 0.05 .
3. **Persistence of excitation:** natural fluctuations or small probing perturbations in $u(t)$ to maintain parameter identifiability.

A practical guideline is that ctDNA sampling every 3–7 days with assay coefficient of variation below 20% is sufficient to track $\chi(t)$ within the adaptive window for the parameter regimes simulated in our computational supplement (to be published separately).

4.2 Estimation methods

Two complementary approaches extract γ and ω from biomarker time series.

4.2.1 Prony method

The Prony method fits exponentially damped sinusoids to a recent data window,

$$z(t) \approx \sum_{k=1}^N A_k e^{-\sigma_k t} \cos(\omega_k t + \phi_k). \quad (15)$$

The dominant mode yields $\gamma \approx \sigma_1$ and $\omega \approx \omega_1$.

4.2.2 Subspace identification (N4SID)

Subspace methods construct Hankel matrices from input–output data and extract a state space realization, directly yielding eigenvalues $\lambda_{\pm} = -\gamma \pm i\omega$.

Both methods can be combined with moving window regression and Kalman filtering to provide continuous $\chi(t)$ updates that are robust to measurement noise.

4.3 Measurement delays and clinical reality

Assay turnaround times (typically 24–72 hours for ctDNA) and pharmacokinetic delays introduce lag between sampling and dose adjustment. The adaptive window $[0.8, 1.0]$ provides sufficient margin to accommodate these delays without destabilizing the control loop. Explicitly accounting for delays in an MPC formulation is straightforward and will be detailed in a methods supplement.

5 Falsification criteria

SymC’s clinical validity rests on explicit, falsifiable predictions. Defined are three independent tests.

5.1 Test 1: χ –window correlation with progression free survival

Prediction: patients who spend more than 70% of monitored time with $0.8 \leq \chi(t) \leq 1.0$ show improved progression free survival (PFS) compared to matched standard of care controls.

Falsification: if prospective Phase II trial data show no significant PFS difference (hazard ratio ≥ 0.9 with $p > 0.05$), the near critical SymC control principle is falsified in this indication.

5.2 Test 2: Prognostic superiority of $\chi(t)$ over raw ctDNA

Prediction: $\chi(t)$ estimated from ctDNA dynamics correlates with relapse events more strongly than raw ctDNA rate of change alone, measured by receiver operating characteristic area under the curve (AUC).

Falsification: if retrospective analysis of existing datasets shows $\text{AUC}_\chi < \text{AUC}_{\text{ctDNA}}$ or $|\text{AUC}_\chi - \text{AUC}_{\text{ctDNA}}| < 0.05$, then $\chi(t)$ fails as a useful prognostic control variable.

5.3 Test 3: Resistance delay in adaptive window

Prediction: maintaining $\chi \in [0.8, 1.0]$ delays emergence of resistant clones compared to continuous high dose therapy, measured by time to progression or genomic sequencing of resistant mutations.

Falsification: if prospective trials show no delay (or accelerated resistance under adaptive control), the substrate synchronization hypothesis underlying multi population SymC control is rejected.

5.4 Cross domain validation

These oncology tests complement independent falsification channels:

- Quantum systems: Lindblad dynamics crossing $\chi = 1$ at a quantum–classical boundary [2].
- Cosmology: $\chi_\delta = 1 \Leftrightarrow q = 0 \Leftrightarrow \Omega_m = 2/3$ identity in flat Λ CDM [3].
- Neurology: basal ganglia oscillations in Parkinson’s disease deviating from a $\chi \approx 0.9$ therapeutic window [4].

Oncology provides a clinically accessible, rapid timescale validation path while cosmological and fundamental physics tests mature over decades.

6 Discussion and translational roadmap

6.1 Substrate inheritance and cancer as χ breakdown

The SymC framework interprets cancer not merely as uncontrolled proliferation but as a loss of inherited near critical substrate structure. Healthy tissues maintain $\chi \approx 1$ through hierarchical feedback: molecular networks inherit χ stability from QCD substrates, cellular metabolism inherits from molecular networks, and tissue homeostasis inherits from cellular regulation. Oncogenesis represents decoupling from these inherited constraints, with cells entering underdamped regimes where growth dominates damping.

Therapy aims to restore near critical dynamics by increasing effective damping γ (drug induced kill rates, immune activation) relative to oscillatory drive ω (proliferation signals, angiogenic feedback). The adaptive window $[0.8, 1.0]$ provides operational guardrails for this restoration process.

6.2 Information efficiency and the $\chi \approx 0.9$ optimum

The choice of $\chi \approx 0.9$ as the practical target, rather than exact $\chi = 1$, follows from information theoretic optimization. In systems with measurement noise, feedback delays, and parameter uncertainty, slight underdamping maximizes the mutual information between control inputs and state observations. This allows adaptive controllers to learn system parameters online while maintaining stability. The trade off between convergence speed (maximized at $\chi = 1$) and parameter observability (improved for $\chi < 1$) is optimized near $\chi \approx 0.9$ under realistic clinical constraints.

6.3 Deployment roadmap

Deployment proceeds through four stages.

6.3.1 Phase 1: Retrospective validation (Months 0–12)

- Apply $\chi(t)$ estimation to existing ctDNA datasets with longitudinal follow up.
- Test correlation with PFS and time to progression.
- Establish observability thresholds and identify optimal cancer types (for example those with high ctDNA shedding and frequent sampling).

6.3.2 Phase 2: Computational optimization (Months 6–18)

- Develop MPC algorithms with toxicity constraints and resistance models.
- Perform Monte Carlo simulations across parameter space.
- Design standardized biomarker pipelines and clinical decision support software.

6.3.3 Phase 3: Prospective Phase II trial (Months 18–48)

- Enroll 60–100 patients in two or three cancer types (for example metastatic prostate, ovarian, colorectal).
- Primary endpoint: PFS with $\chi \in [0.8, 1.0]$ maintained more than 70% of the treatment course.
- Secondary endpoints: time to resistance, quality of life, and treatment free intervals.

6.3.4 Phase 4: Multi site validation and regulatory pathway (Months 48+)

- Expand to multi center trials.
- Integrate with electronic health records and digital health platforms.
- Pursue regulatory approval as a precision medicine diagnostic–therapeutic co development.

6.4 Current limitations

This framework is subject to three primary limitations.

6.4.1 Biomarker quality and sampling frequency

The control law is only as reliable as $\chi(t)$ estimation. Assay turnaround times, measurement noise, and sampling delays bound achievable performance. Cancers with low ctDNA shedding or high biological variability may not provide sufficient signal quality for real time control.

6.4.2 Lack of prospective validation

All inferences remain provisional until Phase II trial data confirm that maintaining the adaptive window improves clinical outcomes. Retrospective analyses provide hypothesis generation, not proof.

6.4.3 Incomplete resistance modeling

This work has outlined the conceptual structure but have not yet fully parameterized multi state models incorporating cancer stem cells, quiescent populations, and resistant clones with distinct χ_k values. Extending CAT–OCT to synchronize control across multiple substrates simultaneously is the subject of ongoing work.

6.5 Broader implications

If validated, CAT–OCT would demonstrate that a universal physical principle, near critical damping as an information efficiency boundary, extends from quantum and cosmological scales to clinical medicine. This would establish SymC as a cross domain organizing principle, with oncology serving as an accessible falsification channel for fundamental physics predictions. The roadmap presented here aims to make that test concrete, quantitative, and achievable within three to five years.

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